Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Aim: To write program for Electric potential difference between two points in free space medium using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:

5-emf (C:\Users\ramac\5-emf) - SciNotes

File Edit Format Options Window Execute ?

*5-emf | 6-emf | emf case study 1

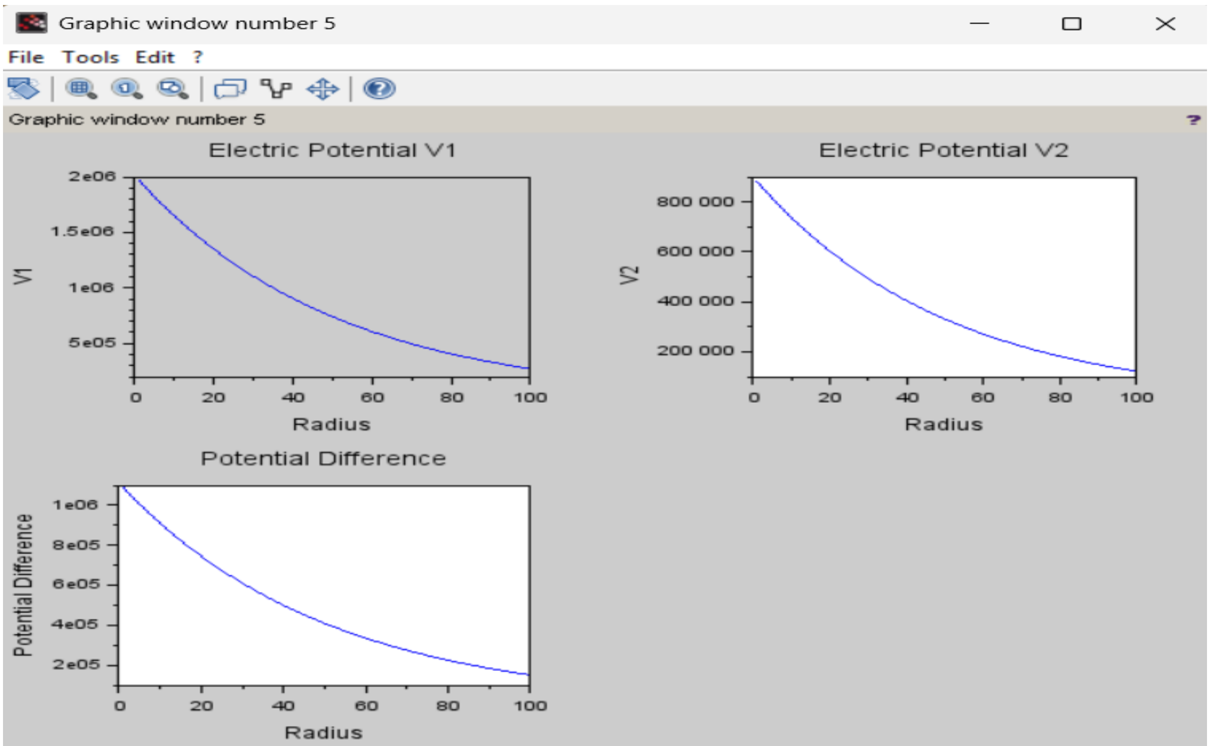
```
clc;
clear;
q = input("Enter the charge q (C): ");
e = 8.854e-12;
pi = 3.14;
ax = input("Enter ax: ");
ay = input("Enter ay: ");
r1 = sqrt(ax^2 + ay^2);
v1 = q / (4 * pi * e * r1);
disp(v1, "Electric Potential at Point A (V1) = ");
bx = input("Enter bx: ");
by = input("Enter by: ");
r2 = sqrt(bx^2 + by^2);
v2 = q / (4 * pi * e * r2);
disp(v2, "Electric Potential at Point B (V2) = ");
v = v1 - v2;
disp(v, "Electric Potential Difference (V) = ");
R = linspace(1,100,100);
Y1 = v1 * exp(-0.02 * R);
Y2 = v2 * exp(-0.02 * R);
Y3 = v * exp(-0.02 * R);
figure();
subplot(2,2,1);
plot(R,Y1);
xlabel("Radius");
ylabel("V1");
title("Electric Potential V1");
subplot(2,2,2);
plot(R,Y2);
xlabel("Radius");
ylabel("V2");
title("Electric Potential V2");
```

Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

```
Enter the charge q (C): 5*10^-6

Enter ax: 0.01

Enter ay: 0.02


   2010739.7

  "Electric Potential at Point A (V1) = "
Enter bx: 0.03

Enter by: 0.04


   899230.12

  "Electric Potential at Point B (V2) = "

   1111509.5

  "Electric Potential Difference (V) = "

-->
```



Theoritical Output:

ax = 1cm   bx = 3 cm
ay = 2 cm   by = 4 cm .

$q = 5 \times 10^{-6}$
r = 3 cm .

① $r_1 = \sqrt{ax^2 + ay^2}$

$\Rightarrow \sqrt{(0.01)^2 + (0.02)^2}$

$\Rightarrow \sqrt{0.0005} \Rightarrow r_1 = 0.02236\,m$

② $v_1 = \frac{q}{4\pi\varepsilon_0 r_1}$ (electric potential at point A

$\Rightarrow \frac{5\times10^{-6}}{4\times3.14\times8.854\times10^{-12}\times0.02236}$

$\Rightarrow \frac{5\times10^{-6}}{2.482\times10^{-12}} \Rightarrow 2.014\times10^{6}\,V.$

③ $r_2 \Rightarrow \sqrt{bx^2 + by^2}$

$\Rightarrow \sqrt{(0.03)^2 + (0.04)^2}$

O/P $\Rightarrow \sqrt{0.0009 + 0.016} \Rightarrow \sqrt{0.0025} \Rightarrow 0.05\,m.$

④ $v_2 \Rightarrow \frac{q}{4\pi\varepsilon_0 r_2}$ (B)

$\Rightarrow \frac{5\times10^{-6}}{4\times3.14\times8.854\times10^{-12}\times0.05} \Rightarrow \frac{5\times10^{-6}}{5.560\times10^{-12}} \Rightarrow 8.993\times10^{5}\,V$

→ electric potential difference.

$v = v_1 - v_2$

$\Rightarrow 2.014\times10^{6} - 8.993\times10^{5}$

$v = 1,115,000\,V.$

Result : Thus the program was verified for Electric potential difference between two points in free space medium using SCILAB was successfully.